Jaehun Kim

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My research interest is in computational design and hardware realization of robotic mechanisms, with a focus on closed-chain linkages and robotic hands.

EDUCATION

Seoul National University (SNU) | Seoul, Republic of Korea **Mar. 2025 – Present**
M.S. in Department of Intelligence and Information
Dynamic Robotic Systems Laboratory (DYROS Lab)
Advisor: Prof. Jaeheung Park
Research area: Computational design of robotic mechanisms, closed-chain linkages, robotic hands
GPA: 4.15/4.3 (Overall) | 4.15/4.3 (Major)

Seoul National University (SNU) | Seoul, Republic of Korea **Mar. 2018 – Aug. 2024**
B.S. in Department of Mechanical Engineering
GPA: 3.80/4.3 (Overall) | 3.89/4.3 (Major)

Sejong Academy of Science and Arts (SASA) | Sejong, Republic of Korea **Mar. 2015 – Feb. 2018**

RESEARCH EXPERIENCES

SNU Dynamic Robotic Systems Lab | Department of Intelligence and Information, SNU **Mar. 2025 – Present**
Graduate Researcher (PI: Prof. Jaeheung Park)

- Topic: Topology-to-Dimension Synthesis of a Structurally Minimal 3-DoF Closed-Chain Robotic Finger
- Conducted research on computational design of robotic hands and closed-chain linkage mechanisms.
- Designed, prototyped, and experimentally validated a palm-actuated linkage-driven robotic finger/hand mechanism.
- Performed kinematic analysis, dimensional optimization, hardware implementation, repeatability test, and fingertip force evaluation.

SNU Healthcare Robotics Lab | Department of Mechanical Engineering, SNU **Mar. 2023 – Dec. 2023**
Undergraduate Researcher (PI: Prof. Amy Kyungwon Han)

- Topic: Haptic Glove with Variable Stiffness
- Proposed new design for VR haptic glove using electrostatic clutches.

SNU Biorobotics Lab | Department of Mechanical Engineering, SNU **Jul. 2022 – Feb. 2023**
Undergraduate Researcher (PI: Prof. Kyu-Jin Cho)

- Topic: Wrist Anchoring System for Exo-Glove Poly

PUBLICATIONS & MANUSCRIPTS

J. Kim, E. Sung, S. Byun, S. You, and J. Park, “Topology-to-Dimension Synthesis of a Structurally Minimal 3-DoF Closed-Chain Robotic Finger,” submitted to IEEE Robotics and Automation Letters.

S. Byun, **J. Kim**, H. Lee, E. Sung, and J. Park, “A lightweight multi-modal haptic glove for high-fidelity grasping state recognition in remote environments,” submitted to International Journal of Control, Automation, and Systems.

S. You, E. Sung, D. Kim, **J. Kim**, and J. Park, "Mechanical Analysis of Plate Harmonic Reducer Considering Deformation of Thin-Plate Flex Spline," International Journal of Precision Engineering and Manufacturing, 1-22, 2026.

E. Sung, S. You, J. Shin, S. Park, S. Hur, **J. Kim**, S. Byun, C. Mun, H. Lee, D. Kim, and J. Park, "SNU-Avatar Haptic Arm: A Hybrid QDD/DD Operator Interface for Teleoperation," IEEE Access, 2026.

WORKING EXPERIENCES

IDeAOcean | Software Researcher **Nov. 2023 – Jun. 2025**

- Contributed to METHEUS, an autonomous mechanism design software platform recognized with a CES 2024 Innovation Award, focusing on algorithmic generation, analysis, and optimization of mechanical systems.
- Developed MATLAB-based software for generating 2D/3D linkage mechanisms from joint-connectivity definitions and randomized geometric parameters, enabling large-scale exploration of candidate mechanism designs.
- Implemented simulation modules to evaluate generated mechanisms through kinematic and dynamic metrics, including end-effector trajectories, actuator torque requirements, joint reaction forces, and force-contour characteristics.
- Designed a layer-assignment algorithm for 2D linkage mechanisms to improve physical realizability by resolving link-overlap and crossing issues through height-level classification.
- Applied multi-objective optimization methods, including NSGA-II, to synthesize industrial linkage mechanisms such as PBV sliding doors and deployable rear-wing systems.

HONORS & AWARDS

Excellence Prize, Korean Maker Star | Korea Institute of Startup and Entrepreneurship Development **Nov. 2023**

Grand Prize, Strange Idea Program | SNU Entrepreneurship Center **Dec. 2023**

MEMBERSHIPS & ACTIVITIES

Republic of Korea Army (Korean Augmentation to the USA) | Republic of Korea **Jul. 2020 – Jan. 2022**

SNU Automotive Club (RunToYou) | Department of Mechanical Engineering, SNU **Sep. 2018 – Aug. 2019**

SKILLS & LANGUAGES

- **Language**: Korean (Native), English (Fluent)
- **Programming & Tools**: MATLAB/Python/ROS 2 (Advanced), C++ (Experienced)
- **Design & Fabrication**: SolidWorks/Fusion 360/3D Printing (Advanced)